



VNiVERSiDAD  
D SALAMANCA

# Teaching Guide and Work- sheets - WE Challenges 2024

All WE are STEAM — Learning expe-  
riences designed by women engineers

Ana B. González-Rogado (coord.) et al.

*Universidad de Salamanca*



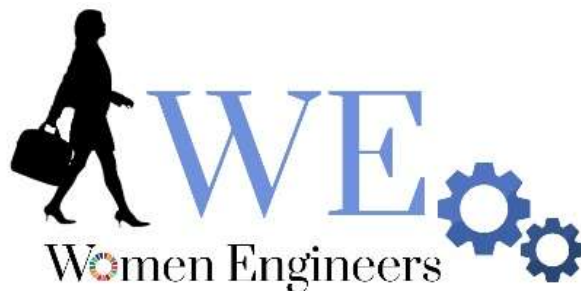
## CONTENTS

|            |                                                                |           |
|------------|----------------------------------------------------------------|-----------|
| <b>I</b>   | <b>Cover</b>                                                   | <b>3</b>  |
| 1          | Cover                                                          | 5         |
| 2          | Index                                                          | 7         |
| 3          | Recommended Citation                                           | 9         |
| 4          | Funding                                                        | 11        |
| <b>II</b>  | <b>Teaching Guide</b>                                          | <b>13</b> |
| 5          | Teaching Guide – WE Challenges 2024                            | 15        |
| 5.1        | Methodological Orientation . . . . .                           | 15        |
| 5.2        | References . . . . .                                           | 16        |
| 5.3        | Acknowledgements . . . . .                                     | 16        |
| <b>III</b> | <b>Challenge Worksheets</b>                                    | <b>17</b> |
| 6          | Challenge Worksheet 1. Km 0 Food                               | 19        |
| 7          | Challenge Worksheet 2. Balance of Flat Figures                 | 21        |
| 8          | Challenge Worksheet 3. Culinary Algorithms                     | 23        |
| 9          | Challenge Worksheet 4. Create the Lever You Need               | 25        |
| 10         | Challenge Worksheet 5. Safe stairs in buildings                | 27        |
| 11         | Challenge Worksheet 6. The “Slime” of Roads                    | 29        |
| 12         | Challenge Worksheet 7. The Planet Is in Our Hands, RECYCLE     | 31        |
| 13         | Challenge Worksheet 8. Accessible Apps                         | 33        |
| 14         | Challenge Worksheet 9. Can We Measure the Comfort of Our Home? | 35        |
| 15         | Challenge Worksheet 10. Secret Messages with a Cipher Disc     | 37        |
| 16         | Challenge Worksheet 11. Circuits with Art                      | 39        |
| 17         | Challenge Worksheet 12. Efficient Facades                      | 41        |

|                                                                                           |           |
|-------------------------------------------------------------------------------------------|-----------|
| <b>18 Challenge Worksheet 13. What Do You Know About “Agri-food By-products”?</b>         | <b>43</b> |
| <b>19 Challenge Worksheet 14. Dog, Cat or Rabbit?</b>                                     | <b>45</b> |
| <b>20 Challenge Worksheet 15. How to Identify if a Facade Is Well Thermally Insulated</b> | <b>47</b> |
| <b>21 Challenge Worksheet 16. How Big Is Sound?</b>                                       | <b>49</b> |
| <br>                                                                                      |           |
| <b>IV Information</b>                                                                     | <b>51</b> |
| <br>                                                                                      |           |
| <b>22 About This Book</b>                                                                 | <b>53</b> |
| <b>23 Licenses</b>                                                                        | <b>55</b> |
| <b>24 How to Cite This Book</b>                                                           | <b>57</b> |
| <b>25 Bibliography</b>                                                                    | <b>59</b> |

Welcome to the **Teaching Guide and Worksheets – WE Challenges 2024** book.

*(Full translation pending)*



- Cover
  - *Cover*
  - *Index*
  - *Recommended Citation*
  - *Funding*
- Teaching Guide
  - *Teaching Guide – WE Challenges 2024*
- Challenge Worksheets
  - *Challenge Worksheet 1. Km 0 Food*
  - *Challenge Worksheet 2. Balance of Flat Figures*
  - *Challenge Worksheet 3. Culinary Algorithms*
  - *Challenge Worksheet 4. Create the Lever You Need*
  - *Challenge Worksheet 5. Safe stairs in buildings*
  - *Challenge Worksheet 6. The “Slime” of Roads*
  - *Challenge Worksheet 7. The Planet Is in Our Hands, RECYCLE*
  - *Challenge Worksheet 8. Accessible Apps*
  - *Challenge Worksheet 9. Can We Measure the Comfort of Our Home?*
  - *Challenge Worksheet 10. Secret Messages with a Cipher Disc*
  - *Challenge Worksheet 11. Circuits with Art*
  - *Challenge Worksheet 12. Efficient Facades*
  - *Challenge Worksheet 13. What Do You Know About “Agri-food By-products”?*
  - *Challenge Worksheet 14. Dog, Cat or Rabbit?*
  - *Challenge Worksheet 15. How to Identify if a Facade Is Well Thermally Insulated*
  - *Challenge Worksheet 16. How Big Is Sound?*
- Information
  - *About This Book*
  - *Licenses*

- *How to Cite This Book*
- *Bibliography*

**Part I**

**Cover**





*(Translation pending)*



*(Translation pending)*



## RECOMMENDED CITATION

*(Translation pending)*



## FUNDING

*(Translation pending)*





# **Part II**

## **Teaching Guide**



## TEACHING GUIDE – WE CHALLENGES 2024

**Authors:** Virginia Aznar-Cuadrado, Ana M.a Vivar-Quintana and Ana B. Gonzalez-Rogado

---

### 5.1 Methodological Orientation

The WE CHALLENGES aim to develop STEAM skills and cover various topics related to the daily lives of students at different educational levels, seeing the usefulness of the content worked on contextually in real-life situations. These challenges can be addressed in the classroom using an active methodology in which students are the protagonists of their own learning.

To face these challenges, they can be presented in a motivating way by placing the students in the role of scientists (engineer, architect...). These challenges can be carried out in small or large groups. Collaborative work is always more enriching, so we suggest forming groups of 3-4 students. These groups should be heterogeneous, made up of boys and girls with different characteristics and abilities so that everyone can contribute their bit to the cooperative work.

Faced with any of these challenges, and depending on the educational level, one can start by encouraging students to make explicit: what are they asking me in the challenge? what do I know about the topic? or what do I need to know? The problem can also be broken down into smaller parts and analyzed progressively.

In the process of solving the challenge, the work of each group can be reviewed and the activity guided with questions that guide the students, especially when designing and validating the required model. The questions can be along the lines of: what do you think might happen if you use/add/change...? why do you think it doesn't work? what elements/materials/tools/variables should you take into account? Or use analogies of processes that are closer or better known to the students.

Also in this process, as in the previous phase, group dynamics techniques can be used, such as Phillips 66 [1-3], or cooperative techniques, such as 1-2-4 [4-6], to share knowledge.

For better organization of the work of each group, the Kanban board [7-8] can be used, simply marking three columns: to do, in progress, and done. In each session, 3-5 minutes are dedicated at the beginning to locate the work done and to be done, decide what will be addressed in the session and distribute tasks, and another 3-5 minutes at the end to review what has been done. This will facilitate the development of each project.

To address the diversity of students in the classroom, tasks can be assigned within each group according to the abilities of each student (design/draw/build the model, search for information, get materials, etc.). Also, to encourage participation without gender bias, roles can be established in the groups (secretary, spokesperson, moderator, coordination, etc.) with well-defined functions that will rotate in each session.

## 5.2 References

- [1] M. Luna Argudín. “Phillips 66”. Accessible from: <https://hadoc.azc.uam.mx/tecnicas/philips.htm>
- [2] A. Barbero (2011), “Dinámicas participativas: Phillips 66”. Accessible from: <https://www.albarbero.com/2011/01/phillips-66.html>
- [3] Universidad Oberta de Cataluña, “Philips 66”. Accessible from: [https://cv.uoc.edu/UOC/a/moduls/90/90\\_156/programa/main/viu/tecniques/viu29.htm](https://cv.uoc.edu/UOC/a/moduls/90/90_156/programa/main/viu/tecniques/viu29.htm)
- [4] J. Blas García (2019), “Técnica cooperativa 4-2-1 como variación de la 1-2-4”. Accessible from: <https://www.jblasgarcia.com/2019/04/tecnica-cooperativa-4-2-1-como.html>
- [5] M. Calvo (2017), “Dinámica cooperativa: uno-dos-cuatro” in HAUTATZEN. Accessible from: <https://hautatzen.net/dinamica-uno-dos-cuatro/>
- [6] Proyecto DIA – Docentes Innovando en el aula (2019), “Técnica 1-2-4”. Accessible from: <https://www3.gobiernodecanarias.org/medusa/ecoblog/mfumfri/2019/10/11/tecnica-1-2-4/>
- [7] P. Santos (2023), “Mejora tu Gestión del Tiempo con Kanban” in PROYECTO MERAKY. Aprender Junt@s SIEMPRE es aprender MÁS. Accessible from: <https://blogsaverroes.juntadeandalucia.es/proyectomeraky/2023/06/01/mejora-tu-gestion-del-tiempo-con-kanban/>
- [8] M. J. Lledó Pardo (2020), “KANBAN para niños” in THiNK productivity. Accessible from: <https://think-productivity.com/kanban-para-ninos/>

## 5.3 Acknowledgements

The Ministry of Equality of the Government of Spain, through the Women’s Institute, funds the WE CHALLENGES action, call PAC 2023 INMUJERES, “ALL WE ARE STEAM – Learning experiences proposed by women engineers (Women Engineers)”, reference number: 31-4ACT/23.

**Part III**

**Challenge Worksheets**



## CHALLENGE WORKSHEET 1. KM 0 FOOD

**Authors:** Ana M.a Vivar-Quintana, Ana B. Ramos-Gavilan y Ana B. Gonzalez-Rogado

---

*(Translation pending)*





## CHALLENGE WORKSHEET 2. BALANCE OF FLAT FIGURES

**Authors:** Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## CHALLENGE WORKSHEET 3. CULINARY ALGORITHMS

**Authors:** Ana B. Gonzalez-Rogado, Ana M.a Vivar-Quintana y Ana B. Ramos-Gavilan

---

*(Translation pending)*



## CHALLENGE WORKSHEET 4. CREATE THE LEVER YOU NEED

**Authors:** Debora Rodrigues, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## CHALLENGE WORKSHEET 5. SAFE STAIRS IN BUILDINGS

**Authors:** M.a Ascension Rodriguez-Esteban, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*





## CHALLENGE WORKSHEET 6. THE “SLIME” OF ROADS

**Authors:** Diana Movilla-Quesada, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## **CHALLENGE WORKSHEET 7. THE PLANET IS IN OUR HANDS, RECYCLE**

**Authors:** Miriam Hernandez-Jimenez, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## CHALLENGE WORKSHEET 8. ACCESSIBLE APPS

**Authors:** Alicia Garcia-Holgado, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## **CHALLENGE WORKSHEET 9. CAN WE MEASURE THE COMFORT OF OUR HOME?**

**Authors:** M. Almudena Frechilla-Alonso, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*





## CHALLENGE WORKSHEET 10. SECRET MESSAGES WITH A CIPHER DISC

**Authors:** Susana Nieto-Isidro, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## CHALLENGE WORKSHEET 11. CIRCUITS WITH ART

**Authors:** Elena Pascual-Corral, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## CHALLENGE WORKSHEET 12. EFFICIENT FACADES

**Authors:** M.a Paz Saez-Perez, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## **CHALLENGE WORKSHEET 13. WHAT DO YOU KNOW ABOUT “AGRI-FOOD BY-PRODUCTS”?**

**Authors:** Isabel Revilla, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*





## CHALLENGE WORKSHEET 14. DOG, CAT OR RABBIT?

**Authors:** Alicia Garcia-Holgado, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## **CHALLENGE WORKSHEET 15. HOW TO IDENTIFY IF A FACADE IS WELL THERMALLY INSULATED**

**Authors:** M.a Soledad Camino-Olea, Ana B. Ramos-Gavilan, Ana M.a Vivar-Quintana y Ana B. Gonzalez-Rogado

---

*(Translation pending)*



## CHALLENGE WORKSHEET 16. HOW BIG IS SOUND?

**Authors:** Ana B. Gonzalez-Rogado, Ana M.a Vivar-Quintana y Ana B. Ramos-Gavilan

---

*(Translation pending)*



# **Part IV**

## **Information**





## ABOUT THIS BOOK

*(Translation pending)*



## LICENSES

This material is distributed under a [Creative Commons Attribution 4.0 International \(CC BY 4.0\)](#) license.

*(Translation pending)*



## HOW TO CITE THIS BOOK

*(Translation pending)*



## **BIBLIOGRAPHY**